Oct. 28

Student Hours This Thursday moved
from 5:45 pm start to 4:15 pm start
still at MLC

Those registered in MATH 440 can do project
and we will use ~~that~~ the 540 scale to determine
course grade.

ch 7 section 16 Neumann Problem

$\Omega \subset \mathbb{R}^n$,

$\partial\Omega$ is nice (can apply divergence Thm)

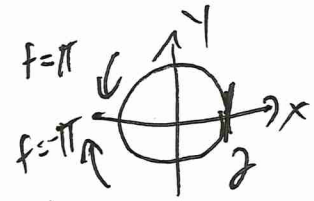
Neumann problem

Find $u \in C^1(\bar{\Omega}) \cap C^2(\Omega)$ so that

$$\Delta u = 0 \quad \text{on } \Omega$$

$$\frac{\partial u}{\partial n} = f \quad \text{on } \partial\Omega$$

where $f \in C^0(\partial\Omega)$ is given.



disk

$\Omega = D(0,2)$

$-\pi \le \theta < \pi$

$f(\theta) = \theta$

not continuous

Uniqueness: None If u is a sol'n, then $u + C$ is also a sol'n. But we can show:

$$\boxed{\text{Thm: Any 2 sol'ns of the Neumann problem differ by a constant}}$$

<u>Pf</u> Let $u_1, u_2 \in C^1(\bar{\Omega}) \cap C^2(\Omega)$ satisfy the N.P. Let $\tilde{u} = u_1 - u_2$

Then $\Delta \tilde{u} = 0$ because $\Delta u_1 = 0$ $\Delta u_2 = 0$

$\frac{\partial \tilde{u}}{\partial n} = 0$ $\frac{\partial u_1}{\partial n} = f$ $\frac{\partial u_2}{\partial n} = f$

Use Green's 1st identity

$$\int_\Omega u \underbrace{\Delta w}_{0} \, dv = \int_{\partial\Omega} u \underbrace{\frac{\partial w}{\partial n}}_{0} \, d\sigma - \int_\Omega \underline{\nabla} u \cdot \underline{\nabla} w \, dv$$

Let $u = w = \tilde{u}$

$$0 = 0 - \int_\Omega \underbrace{\underline{\nabla} \tilde{u} \cdot \underline{\nabla} \tilde{u}}_{\geq 0} \, dv$$

$$\Rightarrow |\underline{\nabla} \tilde{u}|^2 = 0 \Rightarrow \underline{\nabla} \tilde{u} = 0 \underset{HW}{\Longrightarrow} \tilde{u} = \text{const.}$$

Existence

Thm $\int_{\partial\Omega} f \, d\sigma = 0$ is necessary for a sol'n to exist to the N.P.

or If $\int_{\partial\Omega} f \, d\sigma \neq 0 \Rightarrow$ sol'n does not exist.

Pf

$$0 = \int_{\Omega} \Delta u \, dv$$

$$= \int_{\Omega} \underline{\nabla} \cdot \underline{\nabla} u \, dv$$

$$= \int_{\partial\Omega} \underbrace{\underline{\nabla} u \cdot \underline{n}}_{\frac{\partial u}{\partial n}} \, d\sigma$$

$$= \int_{\partial\Omega} f \, d\sigma \quad \blacksquare$$

If the bdry is nice enough, then there is a sol'n (if $\int_{\partial\Omega} f \, d\sigma = 0$). Proof is ~~usu~~ usually by construction.

↪ by giving a sol'n.

So we have uniqueness (up to a constant) + existence (we need $\int_{\partial\Omega} f \, d\sigma = 0$)

contin. dep. on data? Yes.

- Solve by separation of variables $u(x,y) = X(x)\,Y(y)$

- Solve using Fourier series.

Assume $f(\theta)$ has a F-S with coeff

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(\theta)\cos(n\theta)\,d\theta$$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(\theta)\sin(n\theta)\,d\theta$$

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(\theta)\,d\theta = 0$$ for existence

<u>On unit disk</u>

$\Omega = B(0,1) \subset \mathbb{R}^2$

$\Delta u = 0$ in Ω

$\frac{\partial u}{\partial n}(1,\theta) = f(\theta)$ $\quad \theta \in [-\pi, \pi]$

Assume $f \in C^0(\partial\Omega)$ $\quad \int_{-\pi}^{\pi} f(\theta)\,d\theta = 0$

Look for a sol'n of the form

$$u(r,\theta) = \frac{A_0}{2} + \sum_{n=1}^{\infty} \Big[\underbrace{A_n r^n \cos(n\theta)}_{\shortparallel} + \underbrace{B_n r^n \sin(n\theta)}_{\text{are harmonic in } B(0,1)} \Big]$$

satisfies $\Delta u = 0$ in Ω

check B.C.

$$\frac{\partial u}{\partial r}\Big|_{r=1} = \sum_{n=1}^{\infty} \Big[n A_n \cos(n\theta) + B_n n \sin(n\theta) \Big]$$

$$= \sum_{n=1}^{\infty} \Big[a_n \cos(n\theta) + b_n \sin(n\theta) \Big] = f(\theta)$$

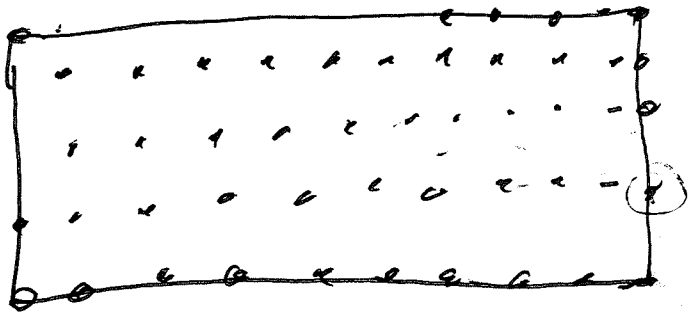
$$A_n = \frac{a_n}{n} \qquad B_n = \frac{b_n}{n}$$

related to avg temp in domain(?)

$$\boxed{u(r,\theta) = \frac{A_0}{2} + \sum_{n=1}^{\infty} \frac{a_n}{n} r^n \cos(n\theta) + \frac{b_n}{n} r^n \sin(n\theta)}$$

arbitrary

Side note In practice, solving a Laplace eqn ~~using~~ numerically:



Prescribe u at every point on $\partial\Omega$: good

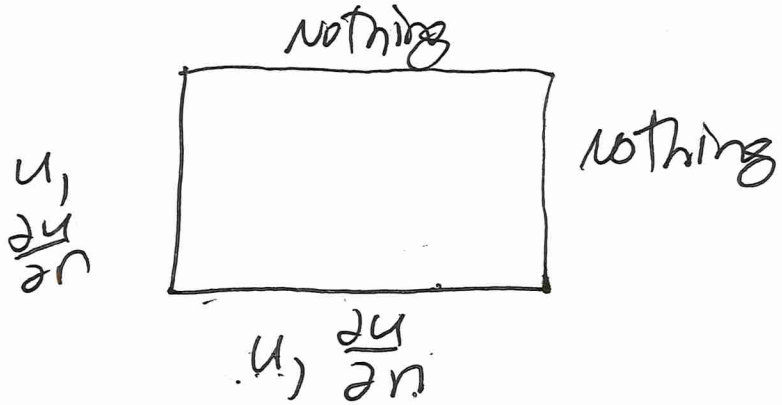
Prescribe $\frac{\partial u}{\partial n}$ " " " " ": Not good

$\Rightarrow A\underline{x} = \underline{b} \Rightarrow$ one eigenvalue of A is zero.

$\Rightarrow A$ is not invertible.

Prescribe $\frac{\partial u}{\partial n}$ some pts + u at other pts (at least one)

Then all is good again :)



not well-posed

$$\alpha u + \beta \frac{\partial u}{\partial n} = f \quad \partial\Omega$$

ok again

mixed BC. given α, β

End of $\Delta u = 0$

Ch 8 Wave Eq'n

Section 1: Sample sol'ns

<u>Wave Eq'n</u> (hyperbolic eq'n)

$$\Delta u - u_{tt} = 0$$

sol'n: $u(x_1, x_2, \ldots, x_n; t)$ $\qquad (\underline{x}, t) \in \mathbb{R}^{n+1}$

spatial ↓ ↙ time

1-Dim wave eq'n: $x \in \mathbb{R}$ (still have 2 indep. variables)

2-Dim wave eq'n: $\underline{x} \in \mathbb{R}^2$ (3 indep. variables)

① Plane waves

Principle part $\Delta u - u_{tt} = 0$

charact. direction $\underline{\xi}$: $\underbrace{(\xi_1)^2 + (\xi_2)^2 + \ldots (\xi_n)^2 - (\xi_t)^2 = 0}_{\text{charact. eq'n}}$

direction is what is important (not magnitude)

Fix ξ_t, $|\xi_t| = 1$ i.e. $\xi_t = \pm 1$

For wave eq'n

$$\xi_1^2 + \xi_2^2 + \dots + \xi_n^2 = 1$$

<u>Def'n</u> The <u>characteristic surface</u> is a surface that has $\underline{\xi}$ as its normal.

Recall the eq'n of a plane

$$n_1 x_1 + n_2 x_2 + n_3 x_3 = d$$

has normal vector $\langle n_1, n_2, n_3 \rangle$.

So $\xi_1 x_1 + \xi_2 x_2 + \dots + \xi_n x_n - t = c$ ← a characteristic surface for $\Delta u - u_{tt} = 0$.

has normal vector $\langle \xi_1, \xi_2, \dots, \xi_n, -1 \rangle$

which is a charact. direction